Lab Exercise 5- Understanding CMD, RUN, and ENTRYPOINT in Dockerfile

Objective:

To learn the differences between CMD, RUN, and ENTRYPOINT instructions in Dockerfiles by creating and running Docker containers with different configurations.

Prerequisites:

- Docker installed on your machine
- Basic understanding of Docker and Dockerfile

Part 1: Overview of CMD, RUN, and ENTRYPOINT

- **RUN:** Executes commands at build time to install software, download dependencies, or configure the environment. The result is saved in the image.
- **CMD:** Specifies the default command to be executed when a container starts. It can be overridden when running a container.
- **ENTRYPOINT:** Defines the main executable for the container, which can't be easily overridden. However, additional arguments can be passed when the container starts.

Part 2: Exploring RUN Command

1. **Create a Dockerfile with RUN:**

Create a directory called dockerfile-run-cmd-entrypoint and navigate to it:

```
mkdir dockerfile-run-cmd-entrypoint && cd dockerfile-run-cmd-entrypoint
```

Create a simple Dockerfile that uses the RUN instruction:

```
# Use an official Ubuntu base image

FROM ubuntu:20.04


# Update the package repository and install curl

RUN apt-get update && apt-get install -y curl


# Print the version of curl

RUN curl --version
```

2. **Build the Docker Image:**

Build the image using the Dockerfile:

```
docker build -t run-example .
```

3. **Explanation:**

The RUN commands in this Dockerfile are executed during the image build process. The first RUN installs curl, and the second RUN command checks and prints the curl version. After the image is built, the commands executed by RUN are already baked into the image.

4. **Verify with Docker History:**

You can check the layers created by RUN using:

```
docker history run-example
```

Each RUN command creates a new layer in the image.

Part 3: Exploring CMD Command

1. **Create a Dockerfile with CMD:**

Modify the Dockerfile to include the CMD instruction:

```
# Use an official Ubuntu base image
FROM ubuntu:20.04

# Install curl
RUN apt-get update && apt-get install -y curl

# Set default command to display the curl version
CMD ["curl", "--version"]
```

2. **Build the Docker Image:**

Build the Docker image again:

```
docker build -t cmd-example .
```

3. **Run the Container:**

Run the container and see the output:

```
docker run cmd-example
```

The output will display the curl version as the default command defined by CMD is executed when the container starts.

4. **Override CMD:**

You can override the CMD by specifying a different command when you run the container:

```
docker run cmd-example echo "Hello from CMD!"
```

This will print Hello from CMD!, showing that the CMD can be overridden at runtime.

Part 4: Exploring ENTRYPOINT Command

1. **Create a Dockerfile with ENTRYPOINT:**

Modify the Dockerfile to use ENTRYPOINT instead of CMD:

```
# Use an official Ubuntu base image
FROM ubuntu:20.04

# Install curl
RUN apt-get update && apt-get install -y curl

# Set entrypoint to curl command
ENTRYPOINT ["curl"]
```

2. **Build the Docker Image:**

Build the image with the ENTRYPOINT instruction:

```
docker build -t entrypoint-example .
```

3. **Run the Container:**

When you run the container, since ENTRYPOINT is set to curl, you need to provide arguments to the curl command:

```
docker run entrypoint-example --version
```

This will print the curl version because ENTRYPOINT defines the main executable (in this case, curl) and --version is passed as an argument to curl.

4. **Override ENTRYPOINT:**

Unlike CMD, the ENTRYPOINT is not easily overridden. If you try to override it using:

```
docker run entrypoint-example echo "Hello from ENTRYPOINT!"
```

It will result in an error because curl will interpret echo as an argument.

However, you can use the --entrypoint option to change the entrypoint:

```
docker run --entrypoint /bin/bash entrypoint-example -c "echo Hello from ENTRYPOINT!"
```

This runs the container with /bin/bash as the entrypoint, overriding the default ENTRYPOINT.

Part 5: Combining CMD and ENTRYPOINT

1. **Create a Dockerfile with Both CMD and ENTRYPOINT:**

Modify the Dockerfile to use both CMD and ENTRYPOINT:

```
# Use an official Ubuntu base image
FROM ubuntu:20.04

# Install curl
RUN apt-get update && apt-get install -y curl

# Set entrypoint to curl
ENTRYPOINT ["curl"]

# Set default arguments to --version
CMD ["--version"]
```

2. **Build the Image:**

Build the new image:

```
docker build -t combined-example .
```

3. **Run the Container:**

When you run the container without specifying any arguments, it will use the CMD as arguments to ENTRYPOINT:

```
docker run combined-example
```

The output will show the curl version, as ENTRYPOINT is curl and CMD provides --version as the argument.

4. **Override CMD Arguments:**

You can override the CMD arguments by specifying your own arguments:

```
docker run combined-example https://www.google.com
```

This command will run curl https://www.google.com inside the container.

Summary of Differences:

- **RUN:** Executes commands during the image build process and creates layers. It is used to install packages and configure the environment.

- **CMD:** Specifies the default command to run when the container starts. It can be overridden by passing a different command when running the container.
- **ENTRYPOINT:** Specifies the main command for the container. It is harder to override but allows passing arguments from the command line. When combined with CMD, CMD provides the default arguments for ENTRYPOINT.

Conclusion:

This lab exercise demonstrates the fundamental differences between RUN, CMD, and ENTRYPOINT in Docker. Each command serves a different purpose, from image build-time configuration (RUN) to defining the container's behavior at runtime (CMD and ENTRYPOINT). Understanding these differences is crucial for building effective and flexible Docker images.

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>mkdir dockerfile-run-cmd-entrypoint

C:\Users\NK>cd dockerfile-run-cmd-entrypoint

C:\Users\NK\dockerfile-run-cmd-entrypoint>echo FROM ubuntu:20.04 > Dockerfile & echo RUN apt-get update ^&^& apt-get install -y curl >> Dockerfile & echo RU
N curl --version >> Dockerfile

C:\Users\NK\dockerfile-run-cmd-entrypoint>docker build -t run-example .
[+] Building 31.2s (7/7) FINISHED                                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                   0.1s
 => => transferring dockerfile: 128B                                                                                                   0.0s
 => [internal] load metadata for docker.io/library/ubuntu:20.04                                                                        4.2s
 => [internal] load .dockerignore                                                                                                      0.1s
 => => transferring context: 2B                                                                                                        0.0s
 => [1/3] FROM docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214                  3.5s
 => => resolve docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214                  0.0s
 => => sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476 27.51MB / 27.51MB                                       2.1s
 => => extracting sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476                                              1.3s
 => [2/3] RUN apt-get update && apt-get install -y curl                                                                               18.2s
 => [3/3] RUN curl --version                                                                                                           0.3s
 => exporting to image                                                                                                                 4.6s
 => => exporting layers                                                                                                                3.5s
 => => exporting manifest sha256:cf8b83cc71cd2c253160f3a6e465aaae68c76326c46b2f4accbcfa106b8aa397                                      0.0s
 => => exporting config sha256:8d2bfb32e18389b6c9acb5138157c96d54547ba3d1ab06a2b4393444b6bab1d1                                        0.0s
 => => exporting attestation manifest sha256:e098466907d743d472b0dd2588a92f130afd2d4305c4618dbfe299e80e64b8a1                          0.1s
 => => exporting manifest list sha256:d37251da54e3dd05bae226a275dd01d2ad13fda0f1403b83ad58c9c4cfd65c16                                 0.0s
 => => naming to docker.io/library/run-example:latest                                                                                  0.0s
 => => unpacking to docker.io/library/run-example:latest                                                                               0.9s

C:\Users\NK\dockerfile-run-cmd-entrypoint>docker history run-exampledocker history run-example
docker: 'docker history' requires 1 argument

Usage:  docker history [OPTIONS] IMAGE

Run 'docker history --help' for more information

C:\Users\NK\dockerfile-run-cmd-entrypoint>echo FROM ubuntu:20.04 > Dockerfile & echo RUN apt-get update ^&^& apt-get install -y curl >> Dockerfile & echo CM
D ["curl", "--version"] >> Dockerfile
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```
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C:\Users\NK\dockerfile-run-cmd-entrypoint>docker build -t run-example .
[+] Building 31.2s (7/7) FINISHED                                                                                                   docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                  0.1s
 => => transferring dockerfile: 128B                                                                                                                  0.0s
 => [internal] load metadata for docker.io/library/ubuntu:20.04                                                                                       4.2s
 => [internal] load .dockerignore                                                                                                                     0.1s
 => => transferring context: 2B                                                                                                                       0.0s
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 => => resolve docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214                                0.0s
 => => sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476 27.51MB / 27.51MB                                                     2.1s
 => => extracting sha256:13b7e930469f6d3575a320709035c6acf6f5485a76abcf03d1b92a64c09c2476                                                             1.3s
 => [2/3] RUN apt-get update && apt-get install -y curl                                                                                              18.2s
 => [3/3] RUN curl --version                                                                                                                          0.3s
 => exporting to image                                                                                                                                4.6s
 => => exporting layers                                                                                                                               3.5s
 => => exporting manifest sha256:cf8b83cc71cd2c253160f3a6e465aaae68c76326c46b2f4accbcfa106b8aa397                                                     0.0s
 => => exporting config sha256:8d2bfb32e18389b6c9acb5138157c96d54547ba3d1ab06a2b4393444b6bab1d1                                                       0.0s
 => => exporting attestation manifest sha256:e098466907d743d472b0dd2588a92f130afd2d4305c4618dbfe299e80e64b8a1                                         0.1s
 => => exporting manifest list sha256:d37251da54e3dd05bae226a275dd01d2ad13fda0f1403b83ad58c9c4cfd65c16                                               0.0s
 => => naming to docker.io/library/run-example:latest                                                                                                 0.0s
 => => unpacking to docker.io/library/run-example:latest                                                                                              0.9s

C:\Users\NK\dockerfile-run-cmd-entrypoint>docker history run-exampledocker history run-example
docker: 'docker history' requires 1 argument

Usage:  docker history [OPTIONS] IMAGE

Run 'docker history --help' for more information

C:\Users\NK\dockerfile-run-cmd-entrypoint>echo FROM ubuntu:20.04 > Dockerfile & echo RUN apt-get update ^&^& apt-get install -y curl >> Dockerfile & echo CM
D ["curl", "--version"] >> Dockerfile
```

```
</script><script nonce="aO_4wgidIjhNl8ex3gPtmw">(function(){window.google.cdo={height:757,width:1440};(function(){var a=window.innerWidth,b=window.innerHeig
ht;if(!a||!b){var c=window.document,d=c.compatMode=="CSS1Compat"?c.documentElement:c.body;a=d.clientWidth;b=d.clientHeight}
if(a&&b&&(a!=google.cdo.width||b!=google.cdo.height)){var e=google,f=e.log,g="/client_204?&atyp=i&biw="+a+"&bih="+b+"&ei="+google.kEI,h="",k=window.google&&
window.google.kOPI||null;k&&(h+="&opi="+k);var l=(new URLSearchParams(window.location.search)).get("gemcid");l&&(h+="&gemcid="+l);var m=(new URLSearchParams
(window.location.search)).get("gemrid");m&&(h+="&gemrid="+m);f.call(e,"","",g+h)};}).call(this);})();(function(){google.xjs={basecomb:'/xjs/_/js/k\x3dxjs.hp
.en.9SA1Dsj5XQI.es5.O/ck\x3dxjs.hp.WupJokhsYfg.L.X.O/am\x3dgAAAAAAAAAAAAAgAAAAAAAAAAAAAAEAAAgAAAQABSQAAAAAAAAAAAAAAAgAAAAAAAAAAACAwAABABCAAAAAEABAAAAAAAAACAAA
ABAAAgAAAAg7ggAAAAAAAABWAQAAAAAAAAAHgAAAAAB/d\x3d1/ed\x3d1/dg\x3d0/ujg\x3d1/rs\x3dACT90oF3XynNep4aL-sva8P13gzj04f7Zg',basecss:'/xjs/_/ss/k\x3dxjs.hp.WupJokh
sYfg.L.X.O/am\x3dgAAAAAAAAAAAAAgAAAAAAAAAAAAAAEAAAgAAAQABSQAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABABCAAAAAEABAAAAAAAAACAAAABAAAg/rs\x3dACT90oFucWZdWLvKhXmA9NbmlYQWv
Jm8OA',basejs:'/xjs/_/js/k\x3dxjs.hp.en.9SA1Dsj5XQI.es5.O/am\x3dAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAAAAAAAgAAAAAAAAAACAwAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAg7ggAAAAAAAABWAQAAAAAAAAAHgAAAAAB/dg\x3d0/rs\x3dACT90oGQOelwcA07I3ziVWMQBw1yEadH-A',excm:[]};})();(function(){var u='/xjs/_/js/k\x3dxjs.hp.en
.9SA1Dsj5XQI.es5.O/am\x3dAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAABAAAAAAAAAAAAAAAAgAAAAAAAAAACAwAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAg7ggAAAAAAAABWAQAAAAAAAA
AHgAAAAAB/d\x3d1/ed\x3d1/dg\x3d4/rs\x3dACT90oGQOelwcA07I3ziVWMQBw1yEadH-A/m\x3dsb_he,d';var st=1;var amd=1000;var mmd=0;var pop=true;var povp=false;var raf=
false;var lfpp=false;var ecb=false;var cst=false;var pxp=false;
var f=this||self;function g(){var a=google.ia&&google.ia.r.B2Jtyd;return a&&[1,5,6].indexOf(a.m)>=0&&a.cbfd&&a.cbvi?a:void 0};function h(a){var b=document.c
reateElement("link");b.as="script";b.href=a;b.rel="preload";document.body.appendChild(b)}function k(){var a=[l];google.dp||(a.forEach(h),google.dp=!0)};goog
le.ps=google.ps||[];function m(a){return a?(a=a.match(/\/ccb=([^/?]+)/))?a[1]:null:null}function p(a){var b=l,d=function(){};google.lx=google.stvsc?d:functi
on(){q(b,a&&raf,a&&lfpp);google.lx=d};google.bx||google.lx()}function r(a,b){b&&(a.src=b);if(!m(b)){var d=a.onload;a.onload=function(c){d&&d(c);google.ps=go
ogle.ps.filter(function(e){return a.src!==e})}}google.ps.push(a.src);document.body.appendChild(a)}
function t(a,b,d,c){d&&(a.fetchPriority="low");b?requestAnimationFrame(function(){r(a,c)}):r(a,c)}google.as=r;function q(a,b,d){google.tick&&google.tick("lo
ad","xjsls");var c=document.createElement("script");c.onerror=function(){google.ple=1};c.onload=function(){google.ple=0};m(a)&&(c.done=function(){google.mum
&&google.mum(a);google.aple=google.aple||0;google.arm&&google.arm[a]&&google.arm[a].resolve()});google.xjsus=void 0;t(c,b,d,a);google.aple=-1;google.dp=!0};
function w(){for(var a=document.getElementsByTagName("img"),b=0,d=a.length;b<d;b++){var c=a[b],e;if(e=c.hasAttribute("data-lzy_")&&Number(c.getAttribute("da
ta-atf"))&1)e=c.getAttribute("jscontroller"),e=!((e===="UBXHI"||e===="R3fhkb"||e===="TSZEqd")&&c.hasAttribute("data-src"));if(e)return!0}return!1};var l,x,y,z,
A,B,C,D,E,F,G="";function H(){l=pxp&&google.xjsup||u;var a=l.match(/\/cb=(loaded_h_\d+)/);a&&a[1]&&(G=a[1]);google.xjsu=l;f._F_jsUrl=l;B=function(d){p(d)};x
=!1;y=(st===1||st===3)&&!!google.caft&&!w();z=g();A=!G&&(st===2||st===3)&&!!z&&!w();if(G){var b=(st===2||st===3)&&!!z;f[G]=function(d){var c=!1,e=function()
{c||(c=!0,cst?setTimeout(function(){return void d.call(window,window._)},0):d.call(window,window._))};b&&google.ia.adls?(z.cbvi.push(function(){delete googl
e.ia.adls}),z.cbvi.push(e),setTimeout(e,mmd)):e()}}C=
pop;D=povp;E=C&&document.prerendering||D&&document.hidden;F=D?"visibilitychange":"prerenderingchange"}function I(a){x||y||A||E||(B(a),x=!0)}
setTimeout(function(){google.tick&&google.tick("load","xjspls");H();if(y||A||E){if(y){var a=function(){y=!1;I()};google.caft(a);setTimeout(a,amd)}A&&(a=func
tion(){A=!1;I()},z.cbvi.push(a),setTimeout(a,mmd));if(E){var b=function(){(D?document.hidden:document.prerendering)||(E=!1,I(!D),document.removeEventListene
r(F,b))};document.addEventListener(F,b,{passive:!0})}x||k()}else B()},0);})();window._ = window._ || {};window._DumpException = _._DumpException = function(
e){throw e};window._s = window._s || {};_s._DumpException = _._DumpException;window._qs = window._qs || {};_qs._DumpException = _._DumpException;window.loa
ded_h_0 = function(cb){cb.call(window,window._);};(function(){var t=[128,0,8388608,0,0,4194304,524288,478154752,2097248,0,64,401408,33185792,548145154,53688
7300,16,276918317,0,335544432,2689597744,2048,599293952,0,4194304,1113,0,122880,4194304];window._F_toggles = window._xjs_toggles = t;})();window._F_installCss
 = window._F_installCss || function(css){};(function(){var pmc='{\x22d\x22:{},\x22sb_he\x22:{\x22client\x22:\x22heirloom-hp\x22,\x22dh\x22:true,\x22ds\x22:\
x22\x22,\x22host\x22:\x22google.com\x22,\x22jsonp\x22:true,\x22msgs\x22:{\x22cibl\x22:\x22Clear Search\x22,\x22dym\x22:\x22Did you mean:\x22,\x22lcky\x22:\x
22I\\u0026#39;m Feeling Lucky\x22,\x22lml\x22:\x22Learn more\x22,\x22psrc\x22:\x22This search was removed from your \\u003Ca href\x3d\\\x22/history\\\x22\\u
100 82106    0 82106    0     0   132k      0 --:--:-- --:--:-- --:--:-- --:--:--  132kx22Search by image\x22,\x22src
h\x22:\x22Google Search\x22},\x22ovr\x22:{},\x22pq\x22:\x22\x22,\x22rfs\x22:[],\x22stok\x22:\x22l0sXiSr5kl_cOzOKHHHY09kzueE\x22}}';google.pmc=JSON.parse(pmc
);})();</script></body></html>
C:\Users\NK\dockerfile-run-cmd-entrypoint>
```